

Towards Building General Purpose Embedding Models for Industry 4.0 Agents

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<https://github.com/IBM/FailureSensorIQ>



Motivation: Embedder for Industry 4.0

Embedders exist in other domains:

💊 Medicine → Disease classification, drug interactions

📈 Finance → Sentiment, fraud detection

However, no embedder for Industry 4.0:

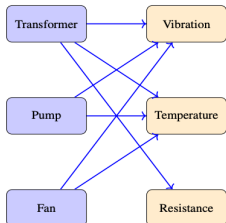
🏭 We define 9 important tasks for Industry and train a multi-task embedder

9 Industrial Tasks

1. Asset → Sensor (A2S), 2. Component → Failure Modes (C2FM), 3. Equipment Class Type → Category (E2CAT), 4. Eq. Category → Class Type (E2CLT), 5. Eq. Unit → Subunits (EU2SU), 6. Failure Mode → Components (FM2CMP), 7. Failure Mode → Class (FM2CLS), 8. Failure Mode → Sensors (FM2S), 9. Sensor → Failure Modes (S2FM)

Example Failure Mode → Sensors (FM2S):

Given an air compressor experiencing a bearing failure, what sensors would detect this fault?



Data Curation

Instruct: What sensors can be applied to detect a fault in an asset and its category?

Query: Asset: electric motor, Category: electric, Fault: stator windings fault

Asset description: Converts electrical energy into mechanical energy to power various industrial machinery.

Fault description: A Stator windings fault is a type of industrial failure mode where the electrical windings in the stator of an electric motor or generator become damaged or degraded, often due to overheating, insulation breakdown, or physical stress, leading to reduced performance, efficiency, or complete failure of the equipment.

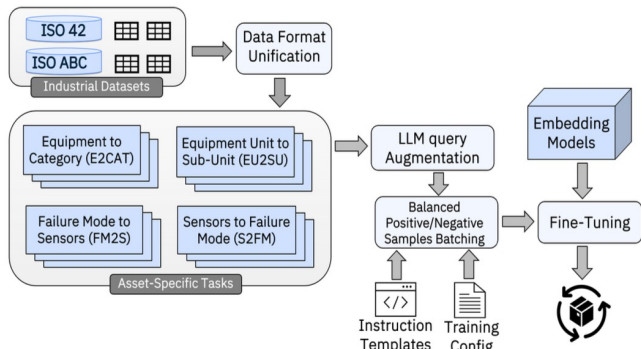
Sensors: Current, Vibration, Temperature, Axial Flux, Cooling Gas

Input, LLM augmented, Ground truth

- We collect ISO tables describing different relationships and prepare instructions for each task
- For each entity (e.g., Asset, Failure Mode) generate with LLM a concise one sentence description

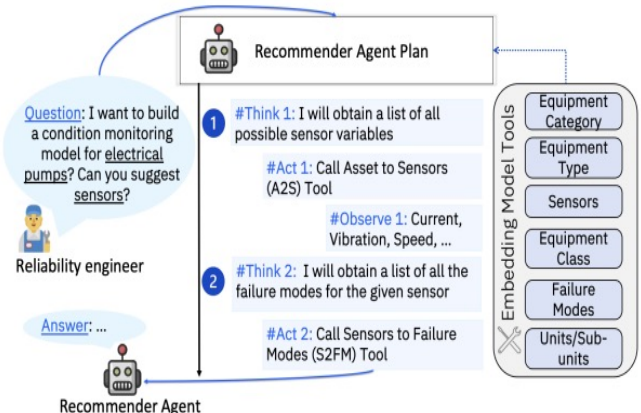
Training Methodology

ISO Documents → Instr. Definition → LLM Augmentation → Batch pos. / neg. balance → Contrastive Learning



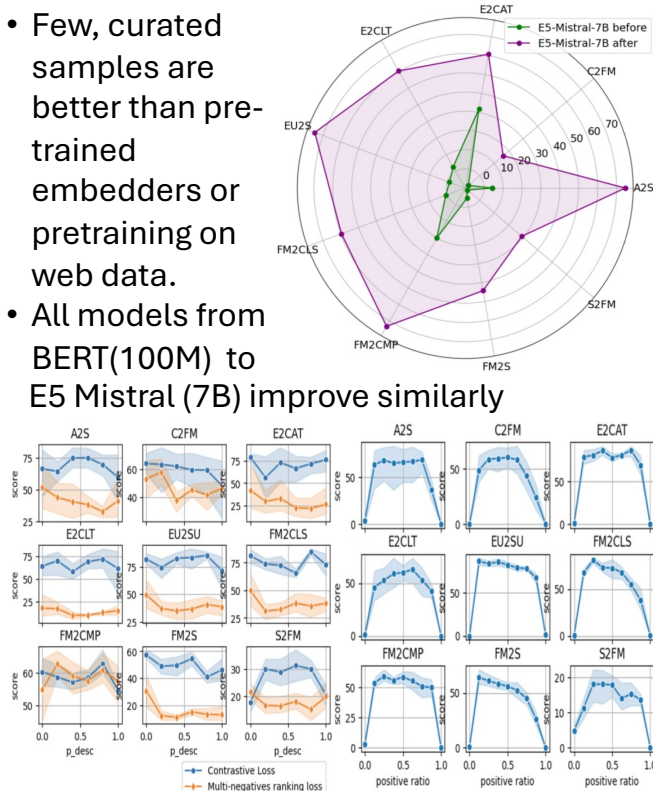
Agent Planning w/ Tools

Agent Planning and Tool Invocation for complex queries using our 9 tools



Findings

- Few, curated samples are better than pre-trained embedders or pretraining on web data.
- All models from BERT(100M) to E5 Mistral (7B) improve similarly



- Balancing in- batch pos./neg. is important
Negative batching can produce in-batch negatives due to dense relation graph
- Query augmentation helps some tasks